Similarity Self-Check Report

(not an official Turnitin report)

SonicAtlas: An Interactive In-Browser Exploratory Data Analysis Platform for Multi-Dimensional Audio Features and Lyrics of Spotify Tracks Across Genres and Eras

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This document records a self-administered similarity check made by the team with open tools. It is NOT a Turnitin or iThenticate report and must not be presented as one. The official similarity check is done by VIT or the course faculty.

1. METHOD

Three complementary checks were run on the final versions of the Project Report, the IEEE paper and every presentation:

- **Fingerprint overlap:** each document was reduced to lower-case word sequences and compared with every source in the corpus. The similarity percentage is the share of a document's words that lie inside an 8-word sequence that also occurs in a source; 5-word overlap is reported as a stricter, noisier view. This is the same idea as the winnowing/fingerprinting used by tools such as MOSS and copydetect.
- **Sentence cosine:** every sentence was compared with every corpus sentence using TF-IDF vectors (unigrams and bigrams); pairs with cosine similarity of 0.6 or more were flagged for manual review.
- **Web exact-phrase search:** a fixed random sample (seed 42) of 60 report sentences and 30 paper sentences was searched on the web as exact quoted phrases.

Excluded from scoring, as the brief requires: the reference lists, URLs and DOIs, numbers, and the titles of cited works (titles are citations, not prose). Standard method and dataset names remain in the text and are counted.

1.1 Corpus and tools

The corpus holds 57 non-empty documents with 93,391 words: the abstracts of the cited papers (retrieved from Crossref, OpenAlex or arXiv), the three Kaggle dataset descriptions, and 33 Wikipedia articles on the methods used (EDA, PCA, t-SNE, LLE, UMAP, imputation, Isolation Forest, ANOVA, chi-square, STL, sentiment analysis, DuckDB and others). Two abstracts could not be retrieved (Brand et al. 2019; Fell and Sporleder 2014) and two Wikipedia pages returned no text. The faculty Sample Report was not available in the project folder, so it could not be added to the corpus. Full texts behind paywalls were not used.

Tool	Version / note
Python	3.13.15
scikit-learn (TF-IDF, cosine)	1.9.1
PyMuPDF (PDF text)	2.0
python-pptx (slide text)	1.0.2
n-gram matcher	docs_build/plagiarism_check.py (this project)
Web search	WebSearch tool (standard mode)

TABLE I: Tools used for the self-check

2. RESULTS

TABLE II gives the final similarity per document. The targets were an overall similarity below 10% and no single source above 2%.

Document	Words checked	Similarity (8-gram)	5-gram overlap	Largest single source	Flagged sentences
Project Report	8,820	0.18%	1.01%	0.09%	0
IEEE Paper	1,740	0.00%	0.00%	0.00%	0
Review2 PPT	1,088	0.00%	0.00%	0.00%	0
Review3 PPT	887	0.00%	0.00%	0.00%	0
Final PPT	1,003	0.00%	0.00%	0.00%	0
Review1 PPT	789	0.00%	0.00%	0.00%	0

TABLE II: Final similarity per document

2.1 Top matched sources and passages

Project Report. Sources with 8-word overlap: paper_wongsuphasawat2016voyager (0.09%), wiki_Music_information_retrieval (0.09%).

> "international conference on acoustics speech and signal processing"

> "voyager exploratory analysis via faceted browsing of visualization"

IEEE Paper. Sources with 8-word overlap: none.

Review2 PPT. Sources with 8-word overlap: none.

Review3 PPT. Sources with 8-word overlap: none.

Final PPT. Sources with 8-word overlap: none.

Review1 PPT. Sources with 8-word overlap: none.

All remaining 8-word matches are proper names of conferences or institutions (for example the name of the IEEE ICASSP conference inside a citation, or the school and university name), which are standard terms and need no rewriting.

2.2 Cross-document check

The IEEE paper was written separately from the report. Only 3.97% of the paper's words fall inside 8-word sequences shared with the report; the shared passages are the affiliation line and the list of cluster names:

> "sonicatlas an interactive in-browser exploratory data analysis platform for multi-dimensional audio features and lyrics of spotify tracks across genres and eras"

“school of computer science and engineering vellore institute of technology vellore”

“school of computer science and engineering vellore institute of technology vellore”

“live recordings upbeat danceable pop soft acoustic songs loud intense rock metal quiet acoustic instrumentals spoken word rap”

2.3 Web exact-phrase check

90 sentences were searched as exact phrases (60 from the report and 30 from the paper). Matches found: report 1, paper 0; 1 sentence made only of numbers could not be searched as a phrase.

Match: "A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text Proc" - https://ojs.aaai.org/index.php/ICWSM/article/view/14550, https://doi.org/10.1609/icwsm.v8i1.14550

The matched query is the title of the VADER paper inside a bibliography entry of the report, so it is a citation and not copied prose. Close paraphrases of abstracts that appeared in search summaries shared no 8-word sequence with the report and are counted as zero.

#	Doc	Exact-phrase query	Matches
1	report	"dance hits from lower-ranked entries reasonably well"	0
2	report	"Background Digital music services now hold catalogues of tens of millions of recordings"	0
3	report	"Quantify the main correlations and visualise genre fingerprints with parallel coordinates	0
4	report	"analytics layer and no in-browser deployment which SonicAtlas has to add"	0
5	report	"This tutorial-style overview reviews the many systems built over the past decade to ease"	0
6	report	"A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text Proc"	1
7	report	"songs the authors test whether feature-space typicality predicts chart performance beyond	0
8	report	"recognition systems in different ways and argues for careful rather than blind use of"	0
9	report	"With k"	error
10	report	"Musicologists and data scientists have used such descriptors to ask how popular music has	0
11	report	"Because the audio-features endpoint is no longer open to new applications these archived	0
12	report	"It is the closest prior study to SonicAtlas tracking acoustic descriptors across three de	0
13	report	"It then surveys adaptations of PCA for different data types and structures and discusses"	0
14	report	"Voyager complements manual chart building with a browsable gallery of automatically gener	0
15	report	"Vega-Lite's declarative charts and selections are a natural specification layer for Sonic	0
16	report	"It was then joined to the main table by Spotify track ID"	0
17	report	"Energy is tightly tied to loudness"	0
18	report	"ANOVA with Tukey HSD Kruskal-Wallis chi-square with Cramer's"	0
19	report	"Findings are reported as static figures for one market without lyrics content or an"	0
20	report	"neighbours UMAP was added as a bonus"	0
21	report	"because they pile up near both"	0
22	report	"Principal Component Analysis and SVD PCA was fitted on all"	0
23	report	"for the whole catalogue for eight representative genres and for every decade from the"	0
24	report	"Embeddings depend on hyperparameters and random seeds and are still not distance-faithful	0
25	report	"with basic and temporally aware audio features and compare several classifiers"	0
26	report	"Bangalore India paper submission deadline"	0
27	report	"MICE validation on artificially masked values"	0
28	report	"An IEEE Bangalore Section conference on computing and communication technologies suitable	0
29	report	"were acoustic and calm"	0
30	report	"row total x column total"	0
31	report	"PROJECT DESCRIPTION AND GOALS"	0
32	report	"Number of canonical tracks per release year Yearly means with"	0

#	Doc	Exact-phrase query	Matches
33	report	"ARIMA forecast of yearly loudness against held-out years Fig"	0
34	report	"Compared with three CNN designs at matched parameter counts the hybrid model gave strong"	0
35	report	"Radar fingerprints of mean audio profiles per genre Fig"	0
36	report	"a review and recent developments Philosophical Transactions of the Royal Society"	0
37	report	"filtering latent factors directly from audio comparing bag-of-words audio features with d	0
38	report	"Lyric sentiment versus audio valence and VADER versus TextBlob score distributions Fig"	0
39	report	"Description of their Work Drawbacks Found in that"	0
40	report	"It is a reference point for using audio descriptors to separate hits from non-hits"	0
41	report	"and average duration shrank from"	0
42	report	"It motivates content-based similarity from audio which SonicAtlas offers as a"	0
43	report	"No Author's Name Title of Paper Journal Name and link Why this paper"	0
44	report	"descriptors are themselves outputs of proprietary models so all conclusions are about the	0
45	report	"The authors present a variant of Stochastic Neighbor Embedding that is easier to optimise	0
46	report	"They span audio-feature analysis genre classification popularity and trend studies the lo	0
47	report	"No Author's Name Title of Paper Journal Name and link Why this paper"	0
48	report	"a research paper a public software artefact and the open documented dataset pipeline behi	0
49	report	"Predicted factors yield sensible recommendations and CNNs clearly outperform the traditio	0
50	report	"CONCLUSION This project set out to understand how the sound of recorded music differs"	0
51	report	"change and diversity across decades directly informing SonicAtlas's"	0
52	report	"Country is the most positive tag"	0
53	report	"exploratory data analysis Spotify audio features dimensionality reduction time series lyr	0
54	report	"Silhouette scores computed with the eight focus genres as labels are slightly negative fo	0
55	report	"around how machine learning is used to recommend or automate the next analysis steps"	0
56	report	"It runs as a server-side Streamlit app in a biomedical domain unlike SonicAtlas's fully"	0
57	report	"It links low-level audio descriptors to perceived emotion which helps interpret Spotify's	0
58	report	"t-SNE is one of the embedding options for SonicAtlas's"	0
59	report	"own laptop with no server which removes the hosting cost that usually stops student"	0
60	report	"and silhouette and name the acoustic clusters found by KMeans"	0
61	paper	"DATASETS AND PREPROCESSING We combine three public Kaggle"	0
62	paper	"VADER and TextBlob agree on polarity for"	0
63	paper	"of the variance in an"	0
64	paper	"Spearman coefficients agree so the relations are monotonic"	0
65	paper	"TABLE II COMPARISON OF DIMENSIONALITY-REDUCTION METHODS Method Time"	0
66	paper	"songs linked to audio descriptors lyric sentiment and valence Fig"	0
67	paper	"can be linked to the main table by track ID or title and artist"	0
68	paper	"All expensive steps run once in Python and emit columnar Parquet files precomputed embedd	0
69	paper	"report strong temporal signals but publish them as static results"	0
70	paper	"shows the yearly trends and Fig"	0
71	paper	"Lyric sentiment is almost unrelated to audio valence"	0
72	paper	"dB while acousticness falls from"	0
73	paper	"t-SNE keeps local neighbourhoods best"	0
74	paper	"are standard for maps of"	0
75	paper	"Exploratory Data Analysis course for their guidance and the creators of the public datase	0
76	paper	"All results are explorable in the browser through SQL executed by DuckDB-WASM on Parquet"	0
77	paper	"with country the most major-key genre Table IV collects the tests"	0
78	paper	"data analysis music information retrieval audio features dimensionality reduction time-se	0
79	paper	"and Min-Max and z-score versions are stored"	0
80	paper	"The closest system in spirit is a Streamlit application for exploratory analysis of clini	0
81	paper	"sample and parallel coordinates Andrews curves radar and violin plots on a balanced sampl	0
82	paper	"bias towards recent releases the low match rate between sources and the English-only lyri	0
83	paper	"s and caught skew and correlation but none flagged the physically impossible tempo values	0
84	paper	"Second results are usually frozen into figures so a reader cannot ask a slightly"	0
85	paper	"Table II compares the four methods and Fig"	0
86	paper	"malised title and lead artist is marked canonical"	0
87	paper	"usually without the matching audio view sentiment is often scored with VADER"	0
88	paper	"Silhouette scores computed with genre labels are negative for every method and in the"	0
89	paper	"English songs the median lyric has"	0
90	paper	"Delivering the results through an in-browser SQL engine lets readers verify and extend ea	0

TABLE III: Web exact-phrase queries and results

2.4 Online checker

Step (c) of the brief, an online checker used without login, was deliberately not run. Free checkers often keep the submitted text in their own databases, so pasting unpublished report text into them could later make the team's own official submission look plagiarised. The decision is left to the team.

3. FIXES AND FINAL NUMBERS

The first run counted the titles of cited works as matches (for example the title of a cited paper written in the literature table). Because titles are citations, they were excluded in the scoring rule; no passage of prose exceeded the threshold, so no text had to be rewritten. TABLE IV shows the effect.

Document	First run (titles counted)	Final (titles excluded)
Project Report	1.40%	0.18%
IEEE Paper	0.00%	0.00%
Review2 PPT	4.70%	0.00%

TABLE IV: Before vs after

Conclusion. The highest overall similarity of any document is 0.18% and the largest share from a single source is 0.09%, both well below the targets (10% overall, 2% per source). These numbers come from the open self-check described above; an official tool with a larger database may report different values. The official Turnitin check is done by VIT or the faculty.